DAY 4 PRACTICE – PYTHON

Exercise 1 – Simple Function

```python
'''
# Exercise 1 - Simple Function
def add_numbers(a, b):
    return a + b

result = add_numbers(5,10)
print(f"sum of two number is : {result}")
```

1] ✓ 0.0s

```
sum of two number is : 15
```

Exercise 2 – Function to Check Even or Odd

```python
'''
# Exercise 2 - Function to Check Even or Odd
def check_even_odd(number):
    if number % 2 == 0:
        print("Even")
    else:
        print("Odd")

check_even_odd(7)
```

[6] ✓ 0.0s

```
Odd
```

Exercise 3 – Function to Find Largest Number

```python
'''
# Exercise 3 - Function to Find Largest Number
def largest_number(lst,n):
    # n = 0
    for i in lst:
        if i > n:
            n = i
    print(f"Largest Number: {n}")

largest_number([10, 45, 2, 78, 33], 0)
```

[11] ✓ 0.0s

```
Largest Number: 78
```

Exercise 4 – Count Words in a Sentence

```python
'''
# Exercise 4 - Count Words in a Sentence
def count_word(word):
    n = word.split()
    print(f"After split method input is: {n}")
    print(f"Number of words: {len(n)}")

word = input("Enter a sentence:")
print(f"Input: {word}")
count_word(word)
```

[13] ✓ 2.8s

```
Input: Python is easy to learn
After split method input is: ['Python', 'is', 'easy', 'to', 'learn']
Number of words: 5
```

Exercise 5 – Write Data to a File

```
# Exercise 5 - Write Data to a File
with open("students.txt", "w") as file:
    file.write("Rahul\n")
    file.write("Amit\n")
    file.write("Priya\n")
    file.write("Neha\n")
```

4] ✓ 0.0s

Exercise 6 – Read Data from a File

```
# Exercise 6 - Read Data from a File
with open("students.txt", "r") as file:
    data = file.readlines()
    for row in data:
        print(row)
```

7] ✓ 0.0s

```
Rahul

Amit

Priya

Neha
```

Exercise 7 – Count Lines in File

```
# Exercise 7 - Count Lines in File
with open("students.txt", "r") as file:
    data = file.readlines()
    print(f"Total number of students: {len(data)}")
```

✓ 0.0s

```
Total number of students: 4
```

Exercise 8 – Append Data to File

```
'''
# Exercise 8 – Append Data to File

with open("students.txt", "a") as file:
    file.write("\nKaran")
    file.write("\nSneha")

with open("students.txt", "r") as file:
    data = file.read()
    print(data)
```

[34] ✓ 0.0s

```
...  Rahul
     Amit
     Priya
     Neha
     Karan
     Sneha
```

Exercise 10 – Find Longest Word

```
'''
# Exercise 10 - Find Longest Word
def longest_words(word):
    n = word.split()
    print(f"Longest word in the sentence is: \'{max(n, key=len)}\'")

word = input("Enter a sentence: ")
print(f"Input: {word}")
longest_words(word)
```

7] ✓ 1.5s

```
Input: Python programming is powerful
Longest word in the sentence is: 'programming'
```

LINUX PRACTICE – DAY 4

Perform the following tasks using terminal:

1.Create a directory called project_data.

2.Inside this directory create files:

students.txt, marks.txt, report.txt

3.Display the list of files inside the directory.

```
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/linux_exercise$ mkdir -p project_data
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/linux_exercise$ ls -lrth
total 4.0K
drwxrwxrwx 1 prashant prashant 4.0K Mar  6 20:20 practice_day1
-rwxrwxrwx 1 prashant prashant  876 Mar  6 20:25 numbers.txt
drwxrwxrwx 1 prashant prashant 4.0K Mar  6 20:42 logs
drwxrwxrwx 1 prashant prashant 4.0K Mar  7 17:32 data_files
drwxrwxrwx 1 prashant prashant 4.0K Mar  8  2026 project_data
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/linux_exercise$ cd project_data/
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/linux_exercise/project_data$ touch students.txt marks.txt report.txt
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/linux_exercise/project_data$ ls -lrth
total 0
-rwxrwxrwx 1 prashant prashant 0 Mar  8  2026 students.txt
-rwxrwxrwx 1 prashant prashant 0 Mar  8  2026 marks.txt
-rwxrwxrwx 1 prashant prashant 0 Mar  8  2026 report.txt
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/linux_exercise/project_data$
```

4.Display the contents of students.txt.

5.Copy students.txt to backup_students.txt.

```
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/linux_exercise/project_data$ cat students.txt
STUDENT_ID NAME   CITY   AGE
S101       Rahul  Pune   20
S102       Sneha  Mumbai 21
S103       Amit   Delhi  22
S104       Priya  Nashik 20
S105       Rohit  Pune   23
S106       Anjali Mumbai 21
S107       Karan  Delhi  22
S108       Meena  Nashik 20
S109       Vijay  Pune   24
S110       Pooja  Mumbai 21
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/linux_exercise/project_data$ ls -lrth
total 0
-rwxrwxrwx 1 prashant prashant 309 Mar  8  2026 students.txt
-rwxrwxrwx 1 prashant prashant 245 Mar  8  2026 marks.txt
-rwxrwxrwx 1 prashant prashant 409 Mar  8  2026 report.txt
-rwxrwxrwx 1 prashant prashant 309 Mar  8  2026 backup_students.txt
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/linux_exercise/project_data$ cat backup_students.txt
STUDENT_ID NAME   CITY   AGE
S101       Rahul  Pune   20
S102       Sneha  Mumbai 21
S103       Amit   Delhi  22
S104       Priya  Nashik 20
S105       Rohit  Pune   23
S106       Anjali Mumbai 21
S107       Karan  Delhi  22
S108       Meena  Nashik 20
S109       Vijay  Pune   24
S110       Pooja  Mumbai 21
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/linux_exercise/project_data$
```

6.Rename report.txt to final_report.txt.

```
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/linux_exercise/project_data$ mv report.txt final_report.txt.
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/linux_exercise/project_data$ ls -lrth
total 0
-rwxrwxrwx 1 prashant prashant 309 Mar  8  2026 students.txt
-rwxrwxrwx 1 prashant prashant 245 Mar  8  2026 marks.txt
-rwxrwxrwx 1 prashant prashant 409 Mar  8  2026 final_report.txt.
-rwxrwxrwx 1 prashant prashant 309 Mar  8  2026 backup_students.txt
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/linux_exercise/project_data$ tree
.
├── backup_students.txt
├── final_report.txt.
├── marks.txt
└── students.txt

1 directory, 4 files
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/linux_exercise/project_data$
```

Commands to explore

Mkdir, cd, touch, ls, cat, cp, mv

SQL PRACTICE – DAY 4

1. Display all employees working in department “Sales”.

[SQL Worksheet]*

```sql
-- 1.  Display all employees working in department "Sales".
SELECT NAME, DEPARTMENT FROM EMPLOYEES3 WHERE DEPARTMENT = 'Sales';
```

Query result | Script output | DBMS output | Explain Plan | SQL history

Download Execution time: 0.004 seconds

	NAME	DEPARTMENT
1	Priya	Sales
2	Pooja	Sales
3	Kavita	Sales

2. Display employees whose salary is between 40000 and 70000.

```sql
--2.   Display employees whose salary is between 40000 and 70000.
SELECT * FROM EMPLOYEES3 WHERE SALARY BETWEEN 40000 AND 70000;
```

Query result | Script output | DBMS output | Explain Plan | SQL history

Download Execution time: 0.006 seconds

	EMP_ID	NAME	DEPARTMENT	SALARY	CITY
1	102	Sneha	Finance	42000	Mumbai
2	103	Amit	IT	60000	Delhi
3	105	Rohit	Marketing	41000	Pune
4	106	Anjali	IT	65000	Mumbai
5	108	Meena	Finance	43000	Nashik
6	109	Vijay	IT	70000	Pune
7	111	Arun	Marketing	45000	Delhi

3. Display employees whose name starts with letter "A".

```sql
-- 3.  Display employees whose name starts with letter "A".
SELECT EMP_ID, NAME FROM EMPLOYEES3;
SELECT * FROM EMPLOYEES3 WHERE NAME LIKE 'A%';
```

Query result | Script output | DBMS output | Explain Plan | SQL history

Download ▾ Execution time: 0.004 seconds

	EMP_ID	NAME	DEPARTMENT	SALARY	CITY
1	103	Amit	IT	60000	Delhi
2	106	Anjali	IT	65000	Mumbai
3	111	Arun	Marketing	45000	Delhi

4. Display number of employees in each department.

```sql
-- 4.  Display number of employees in each department.
SELECT DEPARTMENT, COUNT(*) AS COUNTS FROM EMPLOYEES3 GROUP BY DEPARTMENT;
```

Query result | Script output | DBMS output | Explain Plan | SQL history

Download ▾ Execution time: 0.003 seconds

	DEPARTMENT	COUNTS
1	Sales	3
2	HR	3
3	Marketing	3
4	Finance	3
5	IT	3

5. Display employees from city “Pune”.

```sql
-- 5.   Display employees from city “Pune”.
SELECT * FROM EMPLOYEES3;
SELECT * FROM EMPLOYEES3 WHERE CITY = 'Pune';
```

Query result   Script output   DBMS output   Explain Plan   SQL history

Download   Execution time: 0.003 seconds

	EMP_ID	NAME	DEPARTMENT	SALARY	CITY
1	101	Rahul	HR	35000	Pune
2	105	Rohit	Marketing	41000	Pune
3	109	Vijay	IT	70000	Pune
4	113	Sanjay	Finance	44000	Pune